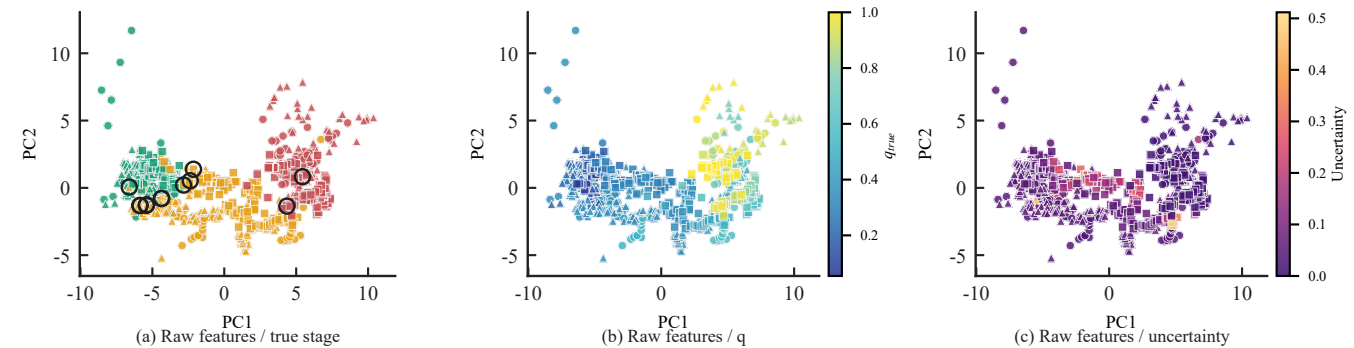
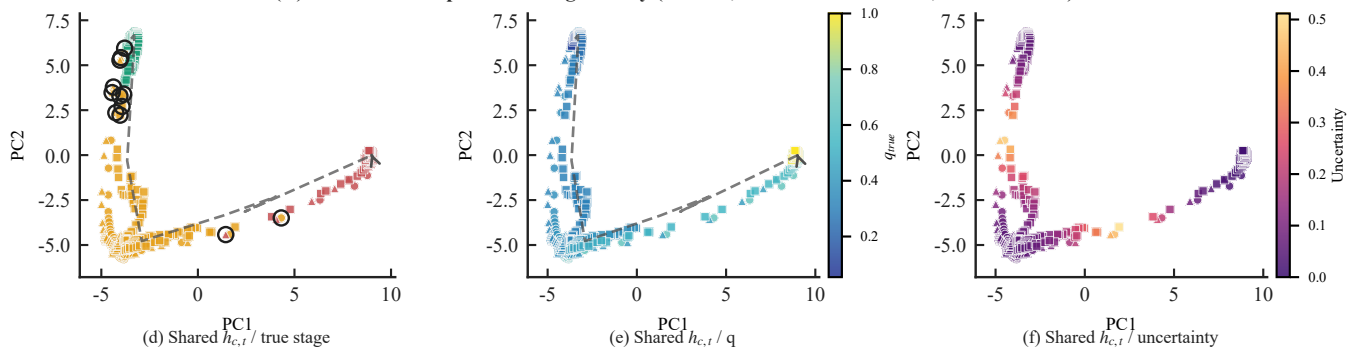
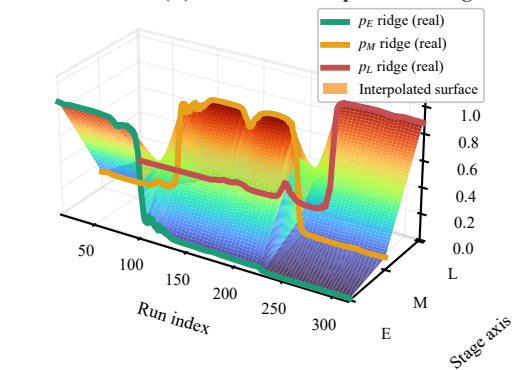




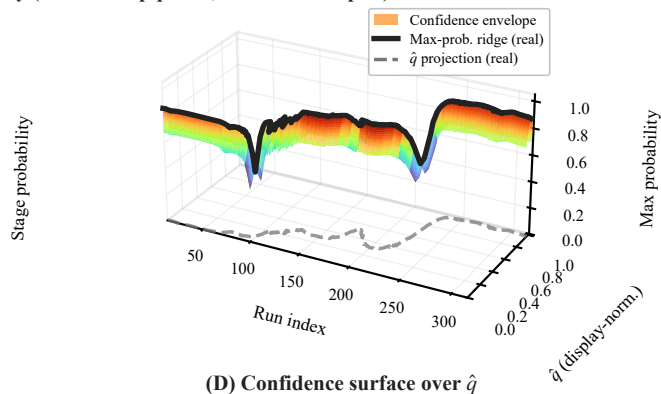
(A) Raw-feature representation geometry (n = 780; C1/C4/C6 conditions, markers o/^/s)



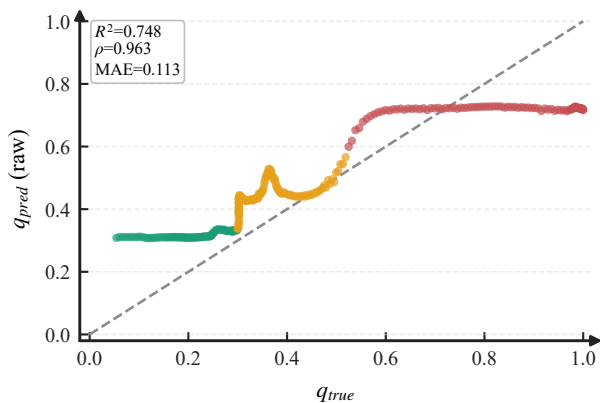
(B) Shared latent representation geometry (same PCA pipeline, same 780 samples)



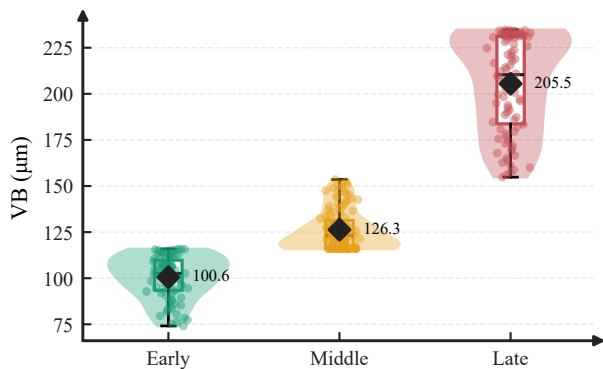
(C) Stage-probability surface over the C6 lifecycle



(D) Confidence surface over $\hat{q}$



(e1) Degradation-position agreement



(e2) Physical wear (VB) by predicted stage